



MATERI HANYA DIGUNAKAN TERBATAS UNTUK PERKULIAHAN
TI5003 PEMODELAN DAN BERPIKIR SISTEM

TI5003 PEMODELAN DAN BERPIKIR SISTEM

PENGANTAR — SILABUS

LUCIA DIAWATI

Study Program of Industrial Engineering and Management

FACULTY OF INDUSTRIAL TECHNOLOGY

INSTITUT TEKNOLOGI BANDUNG



1. LEARNING OUTCOMES (CPMK)

PBS – @DIAWATI 2026

Pada akhir kuliah, mahasiswa:

CPMK 1. Mampu memahami berpikir sistem dan keterkaitannya dengan metodologi sistem dan perkembangannya;

CPMK 2. Mampu mendefinisikan masalah dan sistem revelannya;

CPMK 3. Mampu menerapkan prinsip-prinsip berpikir sistem dalam mendefinisikan sistem relevan dan lingkungannya.

CPMK 4. Mampu menerapkan metodologi sistem untuk pengembangan model dan menggunakannya untuk pengambilan keputusan dalam pemecahan masalah;

CPMK 5. Mampu merancang rencana implementasi dari keputusan atau solusi pemecahan masalah yang dihasilkan.



1. LEARNING OUTCOMES (CPMK)

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Bloom's Taxonomy





2. REFERENSI

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R1	Daellenbach, Hans G. and McNickle, Donald C. Management science Decision making through systems thinking , NY, Palgrave Macmillan, 2005.	Utama
R2	Chevallier, Arnaud. Strategic Thinking in Complex Problem Solving . New York, Oxford University Press 2016.	Utama
R3	Haines, Stephen G. System Thinking and Learning . Massachusetts, HRD Press, 1998.	Pengaya
R4	Gharajedaghi, Jamshid. Systems Thinking: Managing Chaos and Complexity – A Platform for Designing Business Architecture . Amsterdam, Elsevier, 2011.	Pengaya
R5	Marion, Glenn. "An Introduction to Mathematical Modelling." <i>Technical Paper, Bioinformatics and Statistics</i> , Scotlan, University of Bristol, 2008.	Pengaya
R6	Berbagai referensi mengenai Pemodelan dan Berpikir Sistem.	Pengaya



3. MATERI

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No	Topik	Sub-Topik	Refensi
1	System Thinking	System and System Thinking Kompleksitas dalam pengambilan keputusan System thinking vs "Machine Age" Thinking; Elements of System Thinking Feedback System thinking System Methodologies	LD
2	System Concept	Denition of system and its points of views Types of system Relevant system, system boundary and relevant environment Hierarchy of systems Control of system System structure and behaviour	LD
3	Problem Formulation	Problem situation and problem System stakeholders Problem situation mapping tools (rich picture diagram, mind map, cognitive map) Typology of problems	LD



3. MATERI

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No	Topik	Sub-Topik	Refensi Utama
4	Model and System Modeling	System model; Types of model; System modeling process; Conceptual model; Tools for describing a relevant system; Essential properties of good models	LD
5	Soft System Methodology	Soft system paradigm Chekland's Soft system methodology Critical systems heuristics, critical systems thinking, meta-methodologies Model Implementation and Code of Ethics	LD
6-7	Presentasi dan diskusi I:	Paper	Paper
8	UTS	Soal MC dan Essay	
9	Mathematical Modeling (Pemodelan Optimisasi)	Pengertian model matematik Klasifikasi model matematik Tahapan pemodelan matematik The nature of the hard OR process	SPY



3. MATERI

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No	Topik	Sub-Topik	Pengajar
10	Pemodelan Optmisasi	Model Pemrograman Linier dan Pemecahannya	SPY
11		Model Pemrograman Linier Bilangan Bulat dan Pemecahannya	SPY
12		Model Pemrograman Tak Linier dan Pemecahannya	SPY
13		Piranti Lunak untuk Pemecahan Model-model Optimisasi	SPY
14		Contoh-contoh kasus Pemodelan Optimisasi dalam Logistik dan Rantai Pasok	SPY
15-16		Presentasi Tugas Besar	LD/SPY



3. MATERI

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Kuesioner Data Mahasiswa PBS 2026

<https://forms.gle/vx781X8B8FvgDdg66>



4. EVALUASI

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- Ujian Tengah Semester (MC dan Essay) : 30%
- Ujian Akhir Semester (MC dan Essay) : 30%
- Proyek (Tugas Besar) & paper : 30%
- Tugas, Kuis (Tes) : 10%



5. TATA TERTIB

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- Kehadiran minimum 80%;
- Tidak terlambat mengikuti perkuliahan;
- Mengumpulkan semua tugas tepat waktu;
- Mengenakan pakaian yang rapih;
- Tidak menggunakan handphone selama perkuliahan;
- Tidak melakukan kecurangan dalam bentuk apapun.



5. TATA TERTIB

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Academic Honesty

- *Students are expected to be **honest** and **ethical** in their academic work.*
- ***Academic Dishonesty** is defined as an **intentional act of deception** in which a student seeks to claim credit for the work or effort of another person or uses unauthorized materials or fabricated information in any academic work.*



Type of Academic Dishonesty

- *Using ideas, text and other intellectual property developed by someone else while claiming it is your original work is **plagiarism**.*
- *Copying from others or providing answers or information, written or oral, to others is **cheating**.*
- *Unauthorized help from another person or having someone else write one's paper or assignment is **collusion**.*
- *Cheating, plagiarism and collusion are **serious offenses that could result in an E grade and disciplinary action**. Please pay utmost attention to avoid such accusations.*